



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE-3.0

04-07-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

Test Type : SCORE-3.0

Test Pattern : JEE (Advanced)

TEST DATE : 04-07-2021

ANSWER KEY**PAPER-1****PHYSICS**

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,D	A,C	A,C	B,C	A	B,C	A,C	A,C	B,D	A,B,C
SEC-2	Q.	1	2	3	4	5	6				
	A.	A,C	D	A,C	A,B,C,D	A,C,D	A,B,C				
SEC-3	Q.	1	2								
	A.	A->PQS,B->PRS,C->PQ,D->Q	A->RS,B->QRT,C->RS,D->PS								

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	A,B,C,D	A,B	A,B	A,C,D	A,B,C	A,D	A,B,C	A,B,C,D	B,C,D
SEC-2	Q.	1	2	3	4	5	6				
	A.	C,D	A,C	A,B,C	A,C,D	D	C				
SEC-3	Q.	1	2								
	A.	A->PRS,B->PQT,C->P,D->PQT	A->PQR,B->RT,C->ST,D->S								

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A,B	A,D	B	B,D	A,B,C,D	A,B	A,B,C	C,D	C
SEC-2	Q.	1	2	3	4	5	6				
	A.	B,C,D	C,D	B	C	A,B	A				
SEC-3	Q.	1	2								
	A.	A->QRT,B->QS,C->QR,D->PT	A->Q,B->P,C->T,D->PQRST								

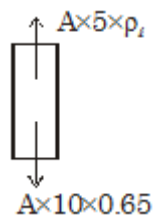
HINT – SHEET

PHYSICS

SEC-1

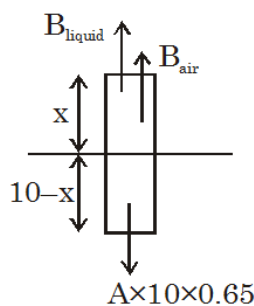
2. Ans (A,C)

Initial



$$S_{\ell} = 1.3$$

final



$$B_{\text{liquid}} = A(10 - x) \times 1.3$$

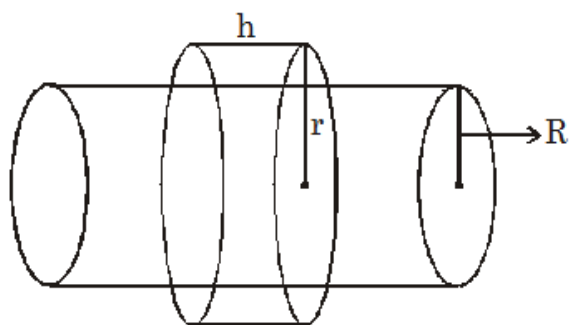
$$B_{\text{air}} = A \times x \times 0.130.$$

$$B_{\text{air}} + B_{\text{liquid}} = mg$$

$$x = \frac{50}{9}$$

$$\text{Displacement } -5 + x = \frac{50}{9}$$

3. Ans (A,C)



$$\frac{\rho \times \pi 2rh}{\epsilon_0} = E \times 2\pi rh$$

$$E_2 = \frac{3R^2}{2\epsilon_0 r} = \frac{4\pi G 3R^2}{2r} = \frac{2\pi G 3R^2}{r}$$

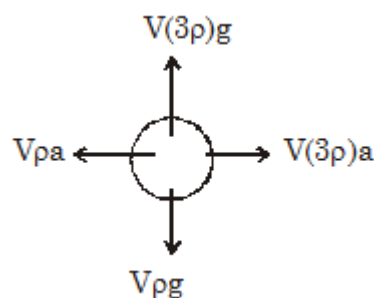
$$E_2 = G \times \frac{\rho \times 4}{3} \frac{\pi R^3}{r^2}$$

$$= \frac{4\pi G \rho R^3}{r} \left(\frac{R}{3r} \right)$$

$$E = G \times \frac{2\pi G \rho R^2}{r} \left(\frac{2R}{3r} \right)$$

HS-2/7 $r \gg R$

6. Ans (B,C)



Since ratio of displacement along y and x is $\frac{1}{2}$

so there must be same relation between acceleration along y-axis and x-axis.

7. Ans (A,C)

$$q = CV$$

$$i = \frac{dq}{dt} = C \frac{dv}{dt}$$

$$V_{CD} = iR$$

for large value of R & C circuit will behave as filter.

8. Ans (A,C)

Surface of conductor is equipotential so potential at point A and B will be same.

9. Ans (B,D)

$$\text{mass rate of strike } \frac{dm}{dt} = A(2v)\rho$$

Change in velocity during strike

$$= \Delta v = \vec{v}_f - \vec{v}_i = (-v) - (v) = -2v$$

$$\text{Thrust force on jet} = \Delta v \frac{dm}{dt} = \frac{dm}{dt} (-2v)$$

$$= -4\rho A v^2$$

Thrust force on wall = - Thrust force on jet =

$$4\rho A v^2$$

$$\frac{d(KE)}{dt} = \frac{1}{2} \frac{dm}{dt} (v_f^2) - \frac{1}{2} \frac{dm}{dt} (v_i^2)$$

$$= \frac{1}{2} \frac{dm}{dt} [(-v)^2 - v^2] = 0$$

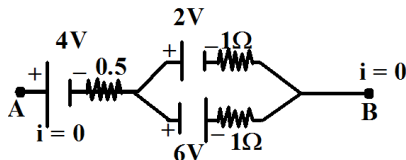
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10. Ans (A,B,C)

$$\text{Internal current } i = \frac{2+6}{1+1} = 4\text{A}$$

$$V_A - 4 - 2 + 1 \times 4 - V_B = 0$$

$$V_A - V_B = 2V$$



PHYSICS

SEC-2

3. Ans (A,C)

$$T = 2\pi\sqrt{\frac{GM}{R^3}}$$

$$T^2 = \frac{4\pi^2 GM}{R^3}$$

$$\frac{2\Delta T}{T} = -\frac{3\Delta R}{R}$$

$$\frac{2 \times 1.08/16}{1.5 \times 3600} = \frac{-3 \times \Delta R}{6400 \text{ km}}$$

$$= \frac{24 \times 3600}{86400} \times 6400 \times 10 = 3\Delta R$$

$$\Delta R = \frac{160}{3} \text{ m}$$

4. Ans (A,B,C,D)

$$V = \sqrt{\frac{GM}{r}}$$

$$\frac{V_A}{V_B} = \sqrt{\frac{r_B}{r_A}} = 2$$

$$E = \frac{-GMm}{2R}, \frac{E_A}{E_B} = \frac{m_A}{m_B} \times \frac{r_B}{r_A} = 12$$

$$u_A = \sqrt{\frac{2GM}{r}}, \quad \frac{u_A}{u_B} = 2$$

$$T = \frac{2\pi}{\sqrt{GM}} (r)^{3/2}$$

$$\frac{T_A}{T_B} = \left(\frac{r_A}{4\pi A} \right)^{3/2} = \frac{1}{8}$$

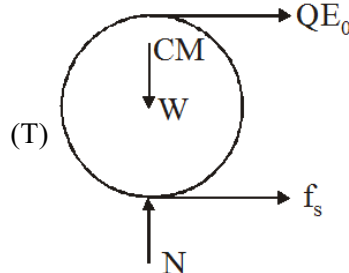
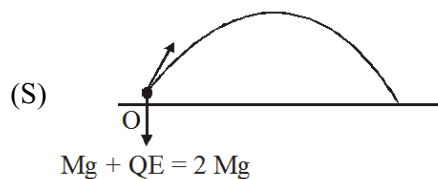
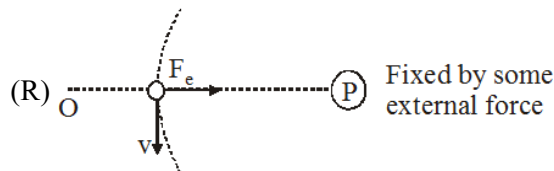
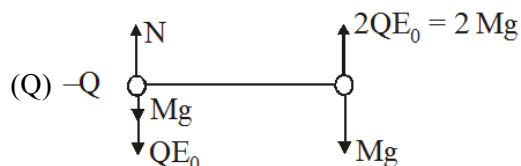
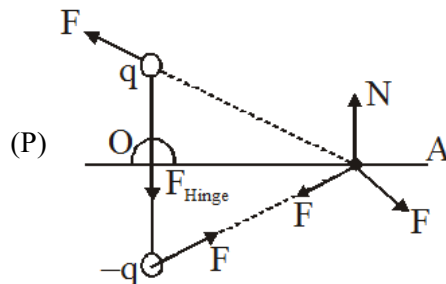
6. Ans (A,B,C)

$$\begin{aligned} \mathbf{E} &= -\frac{\partial \mathbf{v}}{\partial x} \hat{\mathbf{i}} - \frac{\partial \mathbf{v}}{\partial y} \hat{\mathbf{j}} \\ &= -2xy \hat{\mathbf{i}} - x^2 \hat{\mathbf{j}} \end{aligned}$$

PHYSICS

SEC-3

1. Ans (A->PQS,B->PRS,C->PQ,D->Q)



2. Ans (A->RS,B->QRT,C->RS,D->PS)

$$V_{eq} = \frac{\frac{1.5}{2} + \frac{V}{1}}{\frac{1}{2} + \frac{1}{1}}$$

$$= \frac{(V + \frac{3}{4})}{\frac{3}{2}} = \frac{2V}{3} + \frac{1}{2}$$

$$i = \frac{\frac{2V}{3} + \frac{1}{2}}{\frac{2}{3} + 4}$$

CHEMISTRY

SEC-1

1. Ans (B,C,D)

$$\text{Moles of CaO} = \frac{4.48}{56} = 0.08 = \text{moles of CO}_2$$

$$\text{Moles of CaCO}_3 = 0.08$$

3. Ans (A,B)

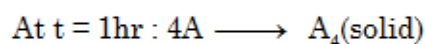


$$t = 0 \quad 8$$

$$t = \infty \quad 0 \quad \frac{8}{x}$$

$$300 = x_B \times 420 = \left(\frac{5}{5 + \frac{8}{x}} \right) \times 420$$

$$x = 4$$



$$(8 - \alpha) \quad \frac{\alpha}{4}$$

$$\text{Total moles} = (8 - \alpha + \frac{\alpha}{4} + 5)$$

$$= (13 - 0.75\alpha)$$

$$360 = \left(\frac{8 - \alpha}{13 - 0.75\alpha} \right) \times 340 + \left(\frac{5}{13 - 0.75\alpha} \right) \times 420$$

$$\alpha = 2$$

4. Ans (A,B)

Correct order of :-

Electron Affinity : $\text{Ne} < \text{Al}$

Ionisation Energy : $\text{Li} < \text{B}$

5. Ans (A,C,D)

During hydrolysis of KHF_2 , H_2 obtained at cathode and F_2 at anode.

6. Ans (A,B,C)

P_4 , S_8 , Cl_2 , Br_2 , I_2 disproportionate in alkali medium.

CHEMISTRY

SEC-2

2. Ans (A,C)

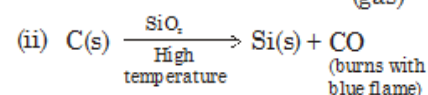
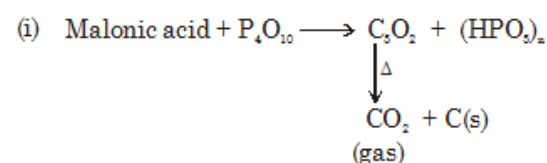
$$9.6 = 9 + \log \left[\frac{\text{Ionized}}{\text{Unionized}} \right]$$

$$\log \left(\frac{\text{Ionized}}{\text{Unionized}} \right) = 0.6 = \log 4$$

$$\% \text{ ionized} = 80\%$$

$$\% \text{ unionized} = 20\%$$

3. Ans (A,B,C)



CO_2 & SO_2 both are acidic, so not distinguish by KOH .

4. Ans (A,C,D)

Oxygen is most abundant in earth crust.

MATHEMATICS

SEC-1

1. Ans (A,C,D)

$$am^2 + 2m + 1 = 0$$

$$m^2 + 2m + a = 0$$

Solving we get $m = \pm 1$.

If $m = 1$ is common root then $a = -3$.

Hence other lines are $y = -\frac{x}{3}$ and $y = -3x$.

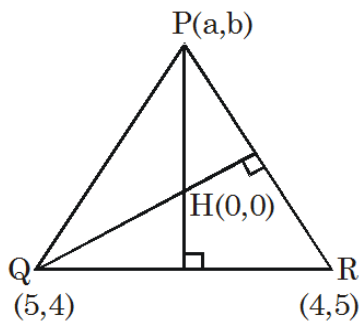
$$\text{Hence } (x + 3y)(3x + y) = 0$$

$$\Rightarrow 3x^2 + 10xy + 3y^2 = 0$$

$$\alpha = 3, \beta = 10, \gamma = 3.$$

2. **Ans (A,B)**

$$M_{PH} \times M_{QR} = -1$$



$$\left(\frac{b}{a}\right)(-1) = -1$$

$$\Rightarrow a = b \quad \dots(1)$$

$$M_{QH} \times M_{PR} = -1$$

$$\frac{4}{5} \times \frac{b-5}{a-4} = -1$$

$$\frac{b-5}{a-4} = -\frac{4}{5} \Rightarrow 4b - 20 = 20 - 5a$$

$$5a + 4b = 40 \quad \dots(2)$$

$$(a, b) = \left(\frac{40}{9}, \frac{40}{9}\right)$$

$$[a + b] = \left[\frac{80}{9}\right] = 8$$

3. **Ans (A,D)**

Unit vector along line L_1

$$\hat{n}_1 = \frac{2\hat{i} - 2\hat{j} - \hat{k}}{3}; \quad \hat{n}_2 = \frac{8\hat{i} + \hat{j} - 4\hat{k}}{9}$$

$$\hat{n}_a = \hat{n}_1 + \hat{n}_2 = \frac{14\hat{i} - 5\hat{j} - 7\hat{k}}{9}$$

$$\hat{n}_b = \hat{n}_1 - \hat{n}_2 = \frac{-2\hat{i} - 7\hat{j} + \hat{k}}{9}$$

$$\vec{r} \cdot \hat{n}_a = 0 \text{ and } \vec{r} \cdot \hat{n}_b = 0$$

4. **Ans (B)**

The general point on line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ is $(\lambda, 2\lambda, 3\lambda)$.

Solve with given plane $x + y + z = 3$ we get

$$\lambda = \frac{1}{2}$$

So point of intersection A with plane is point of intersection with line

$$\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z-3}{-1} \text{ is } B(1, 2, 3)$$

$$\ell_{AB} = \sqrt{\frac{1}{4} + 1^2 + \frac{9}{4}} = \frac{\sqrt{14}}{2} = k$$

$$\therefore 2k = \sqrt{14}$$

5. **Ans (B,D)**

$$\vec{p} \cdot \vec{q} = (x^2\hat{i} - 3\hat{j} + (x+3)\hat{k}) \cdot (\hat{i} + 3\hat{j} - (x-3)\hat{k})$$

$$= x^2 - 9 - (x^2 - 9) = 0$$

$$\Rightarrow \vec{p} \perp \vec{q} \text{ and } |\vec{p}| = |\vec{q}|$$

$$\vec{r} \cdot \vec{s} = (2\vec{p} + 3\vec{q}) \cdot (3\vec{p} - 2\vec{q}) = 0$$

$$\therefore \vec{r} \perp \vec{s}$$

$$\text{i.e. } \vec{r} \wedge \vec{s} = \frac{\pi}{2}$$

6. **Ans (A,B,C,D)**

The points of intersection will be

$$A(\sqrt{5}, 1) \text{ \& } b(-\sqrt{5}, -1).$$

Suppose the 3rd vertex is $C(2\sec\theta, 2\tan\theta)$, then we know that the orthocentre would also lie on the same rectangular hyperbola since Δ is equilateral so orthocentre coincides with centroid which is $\left(\frac{2\sec\theta}{3}, \frac{2\tan\theta}{3}\right)$.

Which clearly doesn't lie on hyperbola $x^2 - y^2 = 4$. Hence no such equilateral triangle can be formed.

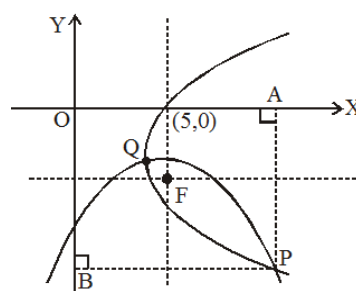
7. **Ans (A,B)**

The area will be maximum when the line $x + y + 2 = 0$ will be diameter of the given circle.

$$\alpha^2 + \beta^2 = 4$$

$$\therefore (a, b) \equiv (2, 0), (-2, 0), (0, 2), (0, -2)$$

8. **Ans (A,B,C)**



Let F be focus.

Now $PA = PF = PB$

So PQ is the common chord & hence $m = -1$

9. **Ans (C,D)**

$$[\vec{n}_1, \vec{n}_2, \vec{n}_3] = 0 \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2a & 1 \\ a & a^2 & 1 \end{vmatrix} = 0$$

$$a = \frac{1}{2}; 1$$

10. Ans (C)

$$\vec{a} \cdot \vec{b} = 3$$

$$|\vec{a}|^2 |\vec{b}|^2 \cos^2 \theta = 9$$

$$\Rightarrow \cos^2 \theta = \frac{9}{5 + 4 \sin 2\alpha} \leq 1$$

$$\therefore \alpha = \frac{\pi}{4} \text{ and } \theta = 0$$

hence $\vec{a}$ and $\vec{b}$ are collinear i.e.

$$\frac{2(\sin \alpha + \cos \alpha)}{\sin \beta} = \frac{1}{\cos \beta} \Rightarrow \tan \beta = 2\sqrt{2}$$

MATHEMATICS

SEC-2

1. Ans (B,C,D)

Solve curves we get

$$y^2 = \frac{144 - 9r^2}{7}$$

$$x^2 = \frac{16r^2 - 144}{7}$$

The equation of tangent to ellipse is

$$y = mx \pm \sqrt{16m^2 + 9}$$

which is also tangent to given circle so

$$r^2 m^2 + r^2 = 16m^2 + 9 \quad \dots\dots(i)$$

Also if A'B'C'D' is square then $m = \pm 1$

$$\therefore r = \frac{5}{\sqrt{2}}$$

2. Ans (C,D)

If A'B'C'D' is a square, then tangents $y = \pm x \pm 5$

for which diagonal length A'B' = 0. Area of

circumcircle of $\Delta A'B'C'$ is 25π .

$$\text{Area of circle } C_1 = \frac{25\pi}{2}$$

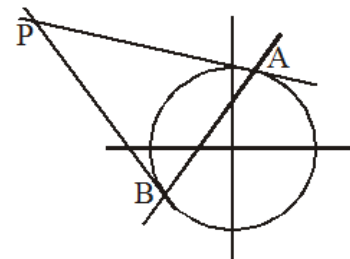
$$\therefore \text{Required ratio} = \frac{1}{2}$$

3. Ans (B)

$$C_1 : x^2 + y^2 = 4$$

$$C_2 : y = x + 1$$

while region R is the disc centered at $(\sqrt{3}, 1)$ and radius



Let $P(x_1, y_1)$

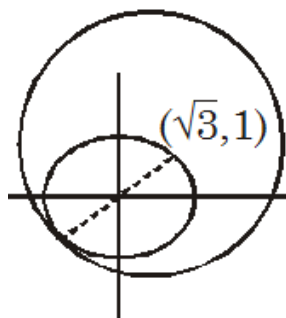
$$\text{Clearly } AB : y - x = 1 \quad \dots\dots(1)$$

is chord of contact

$$\therefore AB : xx_1 + yy_1 = 4 \quad \dots\dots(2)$$

(1) & (2) are identical

$$\frac{x}{-1} = \frac{y_1}{1} = 4 \Rightarrow (x_1, y_1) = (-4, 4)$$



4. Ans (C)

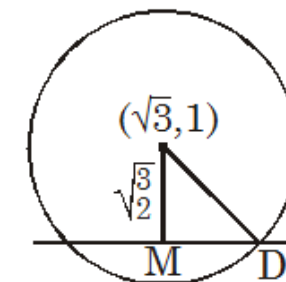
Smallest possible value of 'a' for which

$$A \cap C = A \text{ is } a = 4.$$

$$\therefore R : (x - \sqrt{3})^2 + (y - 1)^2 \leq 4^2$$

$$CD : y = x + 1$$

$$P = \left| \frac{\sqrt{3} + 1 - 1}{\sqrt{2}} \right| = \sqrt{\frac{3}{2}}$$



$$MD = \sqrt{16 - \frac{3}{2}} = \sqrt{\frac{29}{2}}$$

$$\therefore CD = \sqrt{58}$$

5. **Ans (A,B)**

Equation of family of planes containing the line of intersection of P_1 and P_2 is given by $P_1 + \lambda P_2 = 0$

$$\Rightarrow (1 + \lambda)x + (1 - 2\lambda)y + (2 + \lambda)z - 3 - 4\lambda = 0$$

$$\dots\dots(1)$$

Projection of length of line segment AB is greatest, then AB must be parallel to plane (1)

$$\therefore \text{drs of AB} = 2, -1, 3.$$

$$2(1 + \lambda) - 1(1 - 2\lambda) + 3(2 + \lambda) = 0$$

$$\therefore P_3 : 3y + z + 1 = 0.$$

6. **Ans (A)**

Projection of length of line segment is least if AB is perpendicular to plane (1).

$$\frac{1 + \lambda}{2} = \frac{1 - 2\lambda}{-1} = \frac{2 + \lambda}{3} \Rightarrow \lambda = 1$$

$$\therefore \text{Equation of plane } P_4 \text{ is } 2x - y + 3z = 7.$$

MATHEMATICS

SEC-3

1. **Ans (A->QRT, B->QS, C->QR, D->PT)**

$$\text{If } \frac{\beta}{2} = \frac{6}{3} = \frac{8}{4}$$

$\Rightarrow \beta = 4$, then lines are parallel.

If $\beta = 4$ and the point on $L_2(3, 5, 5)$ satisfy L_1 , then the lines are identical otherwise lines are not identical.

$$\therefore \frac{3 - 1}{2} = \frac{5 - 2}{3} = \frac{5 - \alpha}{4} \Rightarrow \alpha = 1$$

If $\beta \neq 4$, then lines are not parallel.

$$D = \begin{vmatrix} \beta & 6 & 8 \\ 2 & 3 & 4 \\ 2 & 3 & 5 - \alpha \end{vmatrix}$$

$$D = (1 - \alpha)(3\beta - 12) = 3(\alpha - 1)(\beta - 4).$$

If $D \neq 0$ lines are skew $\Rightarrow \alpha \neq 1$.

If $D = 0$ lines are intersecting $\Rightarrow \alpha = 1$

2. **Ans (A->Q, B->P, C->T, D->PQRST)**

$$(A) E : \frac{x^2}{2} + \frac{y^2}{3/2} = 1$$

$$b^2 = 2a^2 + \frac{3}{2}$$

$$2a^2 - b^2 = -\frac{3}{2} \Rightarrow \frac{b^2}{3/2} - \frac{a^2}{3/4} = 1$$

$$\frac{y^2}{3/2} - \frac{x^2}{3/4} = 1$$

$$\therefore e = \sqrt{\frac{3}{2}}$$

$$(B) \vec{a} \times \vec{b} = \vec{a} \times (\vec{a} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{a} - \vec{a}^2 \vec{c} = 2\vec{a} - 3\vec{c}$$

$$= 2\vec{a} - 3\vec{c}$$

$$|\vec{a} \times \vec{b} + 3\vec{c} + 2\vec{a}| = |4\vec{a}| = 4\sqrt{3}$$

$$(C) \text{Equation of AB is } S_1 - S_2 = 0$$

$$-10x + 8y = -14$$

$$10x - 8y - 14 = 0$$

$$5x - 4y - 7 = 0$$

$$\therefore \text{Slope} = \frac{5}{4}$$

$$(D) h^2 + k^2 = 1 + 7 = 8$$

Locus of point P is $x^2 + y^2 = 8$

